

DeepSeek-V2: Efficient MoE Language Model with Multi- Head Latent Attention

A Strong, Economical, and Efficient
MoE Language Model

236B Total Parameters | 21B
Activated per Token | 128K Context
Length

Paper:

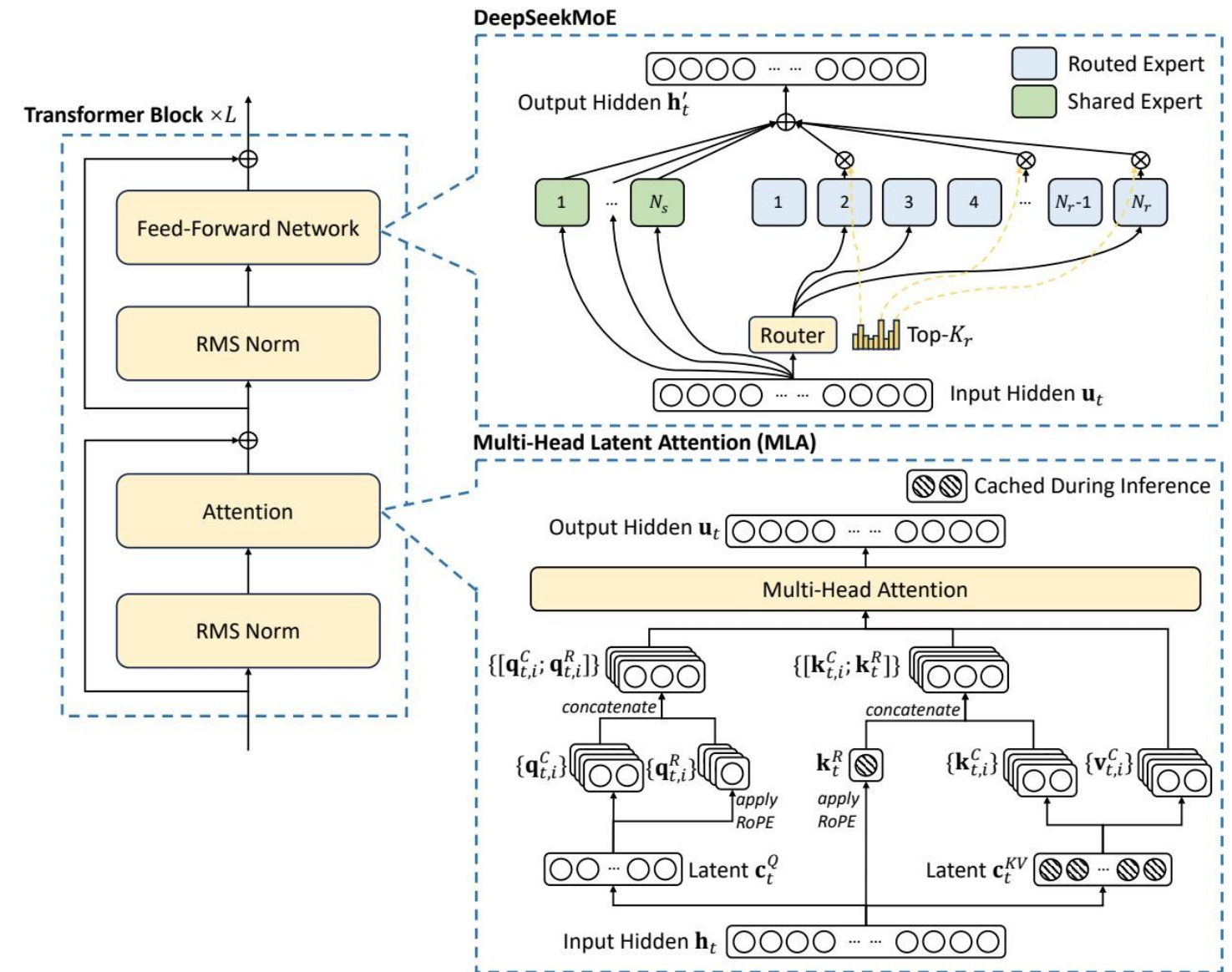
<https://arxiv.org/abs/2405.04434>

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

- Scaling models yields better intelligence but incurs prohibitive training costs.
- Conventional dense models suffer from slow inference speed due to massive compute requirements.
- In autoregressive generation, Multi-Head Attention (MHA) creates a severe KV cache bottleneck, limiting sequence length and batch size.
- A novel Mixture-of-Experts (MoE) architecture coupled with an efficient attention mechanism

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

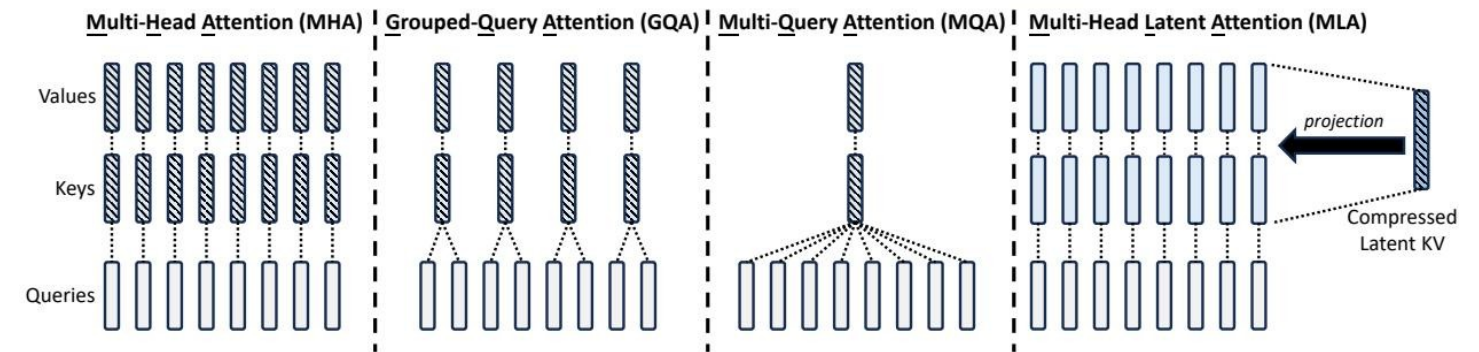
- Transformer block innovations:
- Multi-Head Latent Attention (MLA) replaces standard dense Multi-Head Attention for vastly improved inference efficiency.
- DeepSeekMoE replaces the standard dense Feed-Forward Network (FFN) for improved training and specialized performance.
- Provides a dual optimization for both inference (attention) and training (FFN).



Multi-Head Latent Attention

Compresses KV Cache by 93.3%

- Introduces low-rank key-value joint compression.
- Instead of full K and V vectors, MLA compresses them into a compact latent vector.
- Only the compact latent vector and decoupled RoPE keys are cached during inference.
- Full keys and values are reconstructed on-the-fly via learned projection matrices.
- Achieves better performance than standard MHA while avoiding the accuracy sacrifices of GQA and MQA.



DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

- Shared experts: Always activated, capturing common and general knowledge.
- Routed experts: Sparsely activated via a top-K router. Fine-grained expert segmentation increases specialization.
- Device-limited routing: Constrains each token to experts on at most M devices, minimizing cross-node communication overhead.
- Auxiliary losses: Enforce load balance across experts and devices.
- Token-dropping strategy: Prevents expert

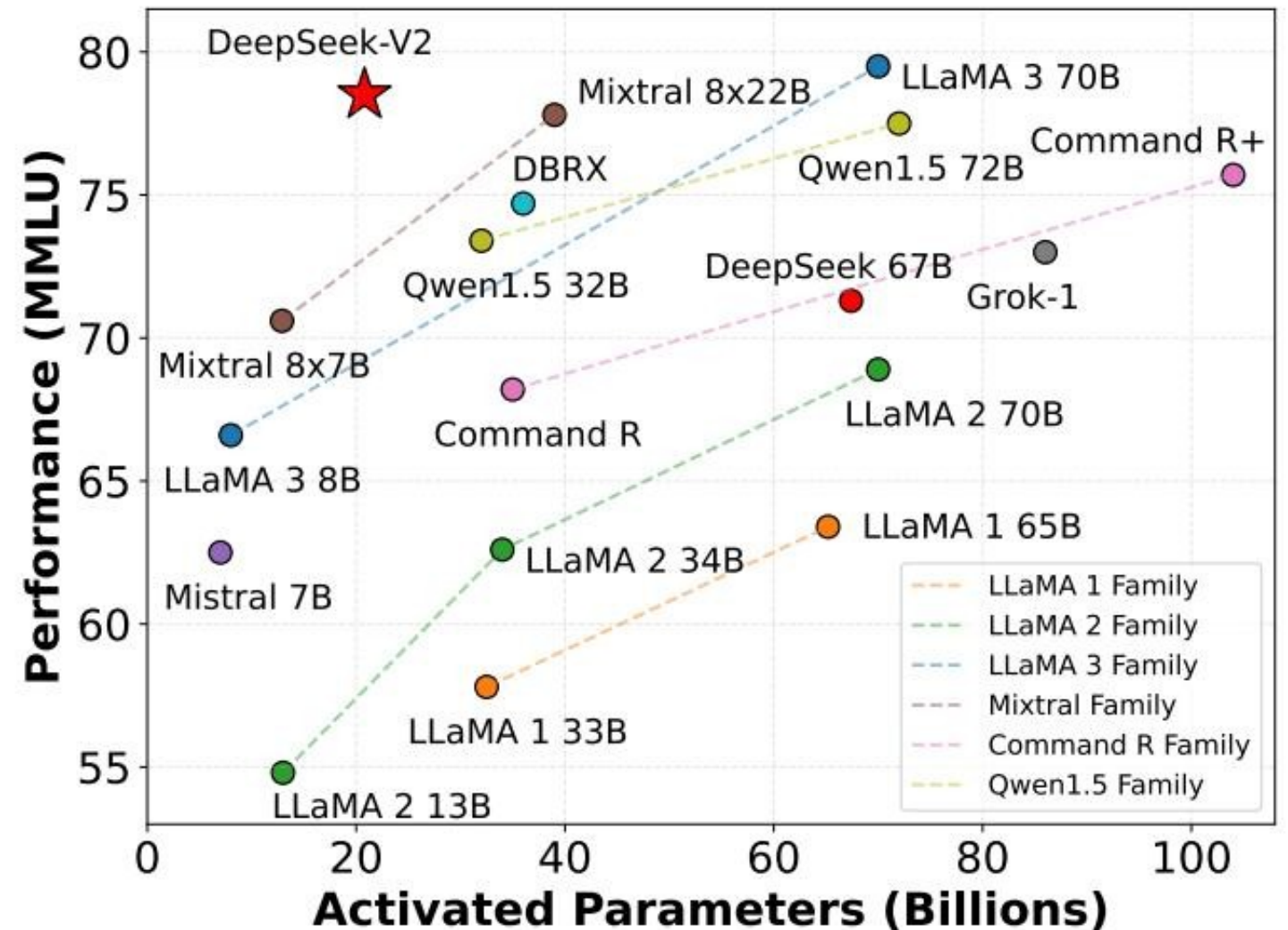
42.5% Training Cost Reduction and 5.76x Generation Throughput

- DeepSeek-V2 relative efficiency gains vs. DeepSeek 67B (Dense Predecessor):
 - Training Cost: 42.5% reduction
 - KV Cache Size: 93.3% reduction
 - Maximum Generation Throughput: 5.76x improvement

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B

Activated Params

- Achieves top-tier MMLU performance (78.5) among open-source models.
- Requires only 21B activated parameters, significantly fewer than competitors.
- Pareto-optimal efficiency compared to LLaMA 3 70B, Mixtral 8x22B, and Qwen1.5 72B.
- Matches or exceeds 70B-class dense models with less than a third of the active parameters.



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

- Post-training pipeline: Supervised Fine-Tuning (SFT) on 1.5M conversational sessions + Reinforcement Learning via Group Relative Policy Optimization (GRPO).
- Evaluation Results:
 - • AlpacaEval 2.0 (length-controlled win rate): 38.9%
 - • MT-Bench (overall): 8.97
 - • AlignBench (overall, Chinese): 7.91
- Outperforms all open-source and most closed-source models in Chinese, demonstrating exceptional multilingual alignment.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

- MLA delivers superior performance to MHA while acting as a strictly better alternative to GQA/MQA for inference efficiency (93.3% KV cache reduction).
- DeepSeekMoE's fine-grained experts and shared expert isolation enable training a 236B-parameter model at 42.5% lower cost than a 67B dense model.
- Efficiency breakthroughs enable matching or exceeding 70B-class dense models on MMLU with only 21B activated parameters.